

Arthur Cavalcante de Andrade

COMPUTER SCIENCE · SOFTWARE ENGINEERING · CYBERSECURITY

Goiânia, Brazil

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Summary

Software Engineer with 4+ years of professional experience, currently working with Python, Ruby on Rails, and PostgreSQL. I design and maintain scalable backend systems and APIs with a strong focus on clean code, performance, and automated testing. Recently, I have focused on architecting advanced AI-driven solutions, leveraging LLMs, RAG architecture, and autonomous AI agents to automate complex workflows and document generation.

My background also includes cybersecurity research, participation in CTFs, and hands-on experience with AppSec and risk management. I am currently part of an IT Residency Program focused on Risk Management and Governance, where I develop secure, compliant software and strategic solutions for the public sector.

I enjoy building solid, secure, and well-documented systems that bridge cutting-edge AI technologies with robust engineering practices to drive business value.

Skills

Operational Systems	Linux, Windows
Databases	PostgreSQL, SQLite.
Languages (Frameworks)	Ruby (Rails), Python (FastAPI), Java (SpringBoot), JavaScript (Vue, React).
Tools & Technologies	Docker, GitHub, GitLab, BitBucket, Burp Suite.
Good Practices	SSDLC, Test-Driven Development (TDD), Algorithm Design, Risk Management.
Languages	Portuguese (Native), English (Proficient)

Work Experience

Court of Justice of the State of Goiás

Goiânia, Brazil - Hybrid

LEAD SOFTWARE ENGINEER

May. 2026 - Present

- Carrying out the technological transition of the MCA contract management and planning system (originally developed in Ruby on Rails and Hotwire) and the Kaizen strategic management system (originally built with Node.js backend and React.js frontend) for the migration and integration of systems into a Java Spring Boot backend;
- Implementing improvements, bug fixes, and optimizations to algorithms and database queries;
- Providing technical guidance to the development team, defining architectural standards, reviewing code, and defining tasks to keep the team aligned with project goals;
- Acting as a bridge between the development team, product managers, and clients to translate business needs into technical solutions.
- Stack: Java (Spring Boot), TypeScript (React.js), PostgreSQL, GitLab.

Court of Justice of the State of Goiás

Goiânia, Brazil - Hybrid

SOFTWARE ENGINEER | IT RISK MANAGER

Feb. 2025 - May. 2026

- Designing and developing risk management systems in a judicial environment, leveraging web frameworks and AI-based solutions to automate workflows and data processing;
- Leading end-to-end risk management processes, including identification, analysis, evaluation, and mitigation across ICT acquisitions, projects, vendors, and emerging technologies;
- Acting as a focal point for external audits, certifications, and third-party assessments, ensuring compliance with regulations and standards, including ISO 31000, LGPD, GDPR, Brazilian Procurement Law (14.133/21), and CNJ Resolution 468/22;
- Developing dashboards and KPIs/KRIs to improve visibility, monitoring, and decision-making;
- Implementing intelligent automation pipelines using AI techniques, reducing manual effort and increasing operational efficiency in risk-related processes.
- Stack: Ruby (Rails 8+, Hotwire), Python (FastAPI), TypeScript (React.js), PostgreSQL, Docker, GitHub.

Cilia Tecnologia (Insurtech)

Goiânia, Brazil - Hybrid

SOFTWARE ENGINEER

Feb. 2023 - Jan. 2025

- Contributing with technical knowledge to ensure project scalability and performance;
- Designing and developing RESTful APIs in Ruby on Rails for high-traffic usage, integrating with external systems and third-party services;
- Updating and optimizing database structures (PostgreSQL), automating tasks, implementing workers, and developing new features and screen flows (Backbone.js and Vue.js);
- Fixing bugs and implemented automated tests (RSpec) to ensure the reliability and functionality of all aspects of the software;
- Leading code reviews and promoting best practices;
- Creating and updating documentations of implemented features;
- Stack: Ruby (Rails 3+), JavaScript (Vue.js, Backbone.js), Python, PostgreSQL, Bitbucket.

National Council for Scientific and Technological Development (CNPq)

Goiânia, Brazil - Hybrid

SCIENTIFIC INITIATION RESEARCHER

Sep. 2022 - Aug. 2023

- Research Project: Study of the induced paired domination number for graphs with bounded maximum degree;
- Scholarship holder in the Institutional Scientific Initiation Scholarship Program (PIBIC);
- Performing studies in graph theory, exploring fundamental concepts, algorithms, and their applications.
- Implementing web scraping algorithms in Python (BeautifulSoup) that represent, analyze, and solve problems in graph structures;
- Analyzing data structures and behavioral properties of induced paired domination for cubic graphs;
- Creating new theorems and conjectures for the research subject.

Education

Postgraduate Degree, IT Processes and Management

Goiânia, Brazil

FEDERAL UNIVERSITY OF GOIÁS

2025 - 2027

Postgraduate Degree, Cybersecurity and Data Governance

Belo Horizonte, Brazil

PUC MINAS

2024 - 2025

B.S., Computer Science

Goiânia, Brazil

FEDERAL UNIVERSITY OF GOIÁS

2018 - 2024

Key Projects

Contracts and Acquisitions Module (MCA)

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COURT OF JUSTICE OF GOIÁS

Apr. 2026

- The Contracts and Acquisitions Module (MCA) is a comprehensive planning and management system implemented at the Court of Justice of Goiás (TJGO) to streamline contract administration and payment workflows. The system successfully transitioned the institution's procurement environment away from scattered spreadsheets by automating processes that were previously handled manually, effectively covering every stage of the procurement lifecycle. Furthermore, the platform features advanced AI-driven modules, including the Risk Assessment System (SAR) for automated contract risk evaluation, and SIMPATIC, which leverages AI agents to automatically generate critical procurement artifacts such as the Demand Formalization Document (DOD), Preliminary Technical Study (ETP), and Terms of Reference (TR).

Risk Assessment Engine

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COURT OF JUSTICE OF GOIÁS | INTELLECTUAL PROPERTY (NO. BR512026002418-8)

Feb. 2026

- This solution features a Python and FastAPI backend engine that automates risk assessment by combining the ISO 31000 methodology with RAG architecture, utilizing natural language processing to transform qualitative data from public sector documents (like DOD, ETP, and TR) into quantitative metrics for calculating inherent and residual risk. By leveraging historical knowledge bases to calibrate probability and impact indices and ensure data traceability, the system serves as a scalable GRC tool across multiple sectors, successfully reducing the total assessment process time by 80%.

Honors & Awards

HACKATHON

- 2023
- Award Winner**, Managed a team of three high school students during a low-code hackathon at the 1st EPT Congress (CETT/UFG), leading the design and development of a mobile app (using MIT App Inventor) integrated with a prototype reservoir to collect and monitor rainwater runoff. The project received a podium award and was later incubated by CEI/UFG.

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